



Reciprocal system of Vectors

If $\vec{a}, \vec{b}, \vec{c}$ & $\vec{a}', \vec{b}', \vec{c}'$ are two sets of noncoplanar vectors such that $\vec{a} \cdot \vec{a}' = \vec{b} \cdot \vec{b}' = \vec{c} \cdot \vec{c}' = 1$ then the two systems are called reciprocal systems.

$$\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]}, \vec{b}' = \frac{\vec{c} \times \vec{a}}{[\vec{a} \vec{b} \vec{c}]}, \vec{c}' = \frac{\vec{a} \times \vec{b}}{[\vec{a} \vec{b} \vec{c}]}$$



Scalar Triple Product/Box Product: $[\vec{a} \vec{b} \vec{c}]$

Box product of $\vec{a}, \vec{b}, \vec{c}$ is $(\vec{a} \times \vec{b}) \cdot \vec{c} = abc \sin \theta \cos \phi$

$\theta \rightarrow$ angle between $\vec{a}$ & $\vec{b}$

$\phi \rightarrow$ angle between $\vec{a} \times \vec{b}$ and $\vec{c}$

Box product geometrically represents the volume of the parallelepiped whose three coterminal edges are represented by $\vec{a}, \vec{b}, \vec{c}$

$$V = [\vec{a} \vec{b} \vec{c}]$$

$$\bullet \vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$\bullet [\vec{a} \vec{b} \vec{c}] = [\vec{b} \vec{c} \vec{a}] = [\vec{c} \vec{a} \vec{b}]$$

$$\bullet \vec{a} \cdot (\vec{b} \times \vec{c}) = -\vec{a} \cdot (\vec{c} \times \vec{b})$$

$$\bullet [\vec{a} \vec{b} \vec{c}] = -[\vec{a} \vec{c} \vec{b}]$$

$$\bullet \text{If } \vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}, \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k},$$

$$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k} \text{ then } [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$\bullet \vec{a}, \vec{b}, \vec{c} \text{ are coplanar if } [\vec{a} \vec{b} \vec{c}] = 0$$

$$\bullet \text{If } \vec{a}, \vec{b}, \vec{c} \text{ are non-coplanar, then } [\vec{a} \vec{b} \vec{c}] > 0 \text{ for right handed system \& } [\vec{a} \vec{b} \vec{c}] < 0 \text{ for left handed system.}$$

$$\bullet (\hat{i} \hat{j} \hat{k}) = 1$$

$$\bullet [k \vec{a} \vec{b} \vec{c}] = k [\vec{a} \vec{b} \vec{c}]$$

$$\bullet [(\vec{a} + \vec{b}) \vec{c} \vec{d}] = [\vec{a} \vec{c} \vec{d}] + [\vec{b} \vec{c} \vec{d}]$$



Direction cosines & Direction Ratios

• If $\vec{a}$ makes angles of α, β, γ with the direction of x, y, z axes, then $\cos \alpha, \cos \beta, \cos \gamma$ are called direction cosines which $\vec{a}$ is usually denoted by l, m, n .

• Any three members a, b, c proportional to the direction cosines of a line are called direction ratios

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$$

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}; m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}$$

$$n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}} \cdot l^2 + m^2 + n^2 = 1$$



Vector Equation of a Line

• Parametric vector equation of a line passing through two points

$$A(\vec{a}) \& B(\vec{b}) \text{ is } \vec{r} = \vec{a} + t(\vec{b} - \vec{a})$$

• If line passes through the point $A(\vec{a})$ & is parallel to

$$\vec{b}, \text{ then its equation is } \vec{r} = \vec{a} + t\vec{b}$$

• Equation of the bisectors of the angle between the lines,

$$\vec{r} = \vec{a} + \lambda \vec{b} \& \vec{r} = \vec{a} + \mu \vec{c} \text{ is } \vec{r} = \vec{a} + t(\vec{b} + \vec{c}) \& \vec{r} = \vec{a} + p(\vec{c} - \vec{b})$$



Shortest distance between two lines

If two lines are $\vec{r}_1 = \vec{a}_1 + k\vec{b}$ & $\vec{r}_2 = \vec{a}_2 + k\vec{b}$

$$\text{then } d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$$



Equation of Plane

$(\vec{r} - \vec{r}_0) \cdot \vec{n} = 0$ containing the point with position vector $\vec{r}_0$, where $\vec{n}$ is a vector normal to the plane.

$\vec{r} \cdot \vec{n} = d$ general equation



Projection & Component

$$\bullet \text{Projection of } \vec{a} \text{ along } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

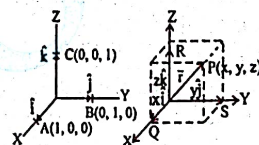
$$\bullet \text{Component of } \vec{a} \text{ along } \vec{b} = \frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$$

$$\bullet \text{Projection of } \vec{a} \perp \vec{b} = \frac{|\vec{a} \times \vec{b}|}{|\vec{b}|}$$

$$\bullet \text{Component of } \vec{a} \perp \vec{b} = \vec{a} - \frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$$



Component Of Vector



$\vec{OA}, \vec{OB}$ & $\vec{OC}$ are unit vectors along x, y & z axes respectively, denoted by $\hat{i}, \hat{j}$ & $\hat{k}$ respectively Position Vector of with reference to O is given by:

$$\vec{OP} \text{ (or } \vec{r}) = x\hat{i} + y\hat{j} + z\hat{k}$$

This form of any vector is called its component form.

$$\text{Also, } |\vec{OP}| = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$



Vector Joining Two Points

Let $A(x_1, y_1, z_1)$ & $B(x_2, y_2, z_2)$ be any two points in the space, then

$$\vec{OA} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k} \& \vec{OB} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$$

$$\therefore \vec{AB} = \vec{OB} - \vec{OA} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

$$|\vec{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



Section Formulae

The position vector of a point R dividing a line segment joining the points P & Q whose position vectors are

$\vec{a}$ & $\vec{b}$ respectively, in the ratio $m : n$

$$(i) \text{ internally, is given by } \frac{m\vec{b} + n\vec{a}}{m + n}$$

$$(ii) \text{ externally, is given by } \frac{m\vec{b} - n\vec{a}}{m - n}$$

The position vector of the middle point of PQ is given by $\frac{1}{2}(\vec{a} + \vec{b})$



Scalar product of four vectors

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{b} \cdot \vec{c} \\ \vec{a} \cdot \vec{d} & \vec{b} \cdot \vec{d} \end{vmatrix}$$



Vector product of four vectors

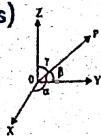
$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \vec{b} \vec{d}]\vec{c} - [\vec{a} \vec{b} \vec{c}]\vec{d}$$

THREE DIMENSIONAL GEOMETRY

1. Direction Cosines of A Line (Dc's)

The direction cosines are generally denoted by l, m, n .

Hence, $l = \cos \alpha, m = \cos \beta, n = \cos \gamma$
Note that $l^2 + m^2 + n^2 = 1$



3. Equation of Line

1. Equation of a line through a given point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$:

In vector form, $\vec{r} = \vec{a} + \lambda \vec{b}$

In cartesian form, $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

where, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$, $\vec{b} = a\hat{i} + b\hat{j} + c\hat{k}$. Here, a, b, c are also the direction ratios of the line.

2. Equation of a line passing through two given points with position vectors $\vec{a}$ and $\vec{b}$:

In vector form, $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

In cartesian form, $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$ where, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$
 $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$
 $\vec{b} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$

2. Direction Ratio of A Line (Dr's)

Any three numbers a, b and c proportional to the direction cosines l, m and n , respectively are called direction ratios of the line.

The direction ratios of a line passing through two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are $(x_2 - x_1), (y_2 - y_1), (z_2 - z_1)$

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$$

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

4. Angle Between Two Lines

In vector form, The angle between two lines $\vec{r} = \vec{a} + \lambda \vec{b}$ & $\vec{r} = \vec{c} + \mu \vec{d}$ is given as:

$$\cos \theta = \frac{|\vec{b} \cdot \vec{d}|}{|\vec{b}| |\vec{d}|}$$

In cartesian form, The angle between two lines:

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \text{ and } \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \text{ is } \cos \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

• If two lines are perpendicular, then $\vec{b} \cdot \vec{d} = 0$ or $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

• If two lines are parallel, then $\vec{b} = \lambda \vec{d}$ or $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

5. Shortest Distance Between Two Lines

1. Distance Between Parallel Lines the shortest distance between parallel lines

$$L_1: \vec{r} = \vec{a}_1 + \lambda \vec{b} \text{ and } L_2: \vec{r} = \vec{a}_2 + \mu \vec{b} \text{ is } d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$$

2. Distance Between Two Skew Lines In vector form, The distance between two skew lines

$$\vec{r} = \vec{a}_1 + \lambda \vec{b}_1 \text{ & } \vec{r} = \vec{a}_2 + \mu \vec{b}_2$$

$$d = \frac{|(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|}$$

In cartesian form,

The distance between two skew lines:

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \text{ and } \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \text{ is:}$$

$$d = \frac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1 c_2 - b_2 c_1)^2 + (c_1 a_2 - c_2 a_1)^2 + (a_1 b_2 - a_2 b_1)^2}}$$

6. Equation of A Plane In Normal Form

Vector Form

$$\vec{r} \cdot \hat{n} = d$$

Here $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\hat{n}$ is the unit vector along the normal from origin to the plane.

d is perpendicular distance of the plane from the origin.

Cartesian Form

$$lx + my + nz = d$$

where l, m, n are the direction cosines of $\hat{n}$ (unit vector along the normal from origin to the plane).

7. Equation Of A Plane Perpendicular To A Given Vector And Passing Through A Given Point

Vector Form

Let a plane pass through a point with position vector $\vec{a}$ and perpendicular to the vector $\vec{N}$. Then its equation is given as: $(\vec{r} - \vec{a}) \cdot \vec{N} = 0$

Cartesian Form

Let a plane pass through a point (x_1, y_1, z_1) & the direction ratio of the vector perpendicular to the plane be A, B, C . Then its equation is given as:

$$A(x-x_1) + B(y-y_1) + C(z-z_1) = 0$$

8. Equation Of A Plane Passing Through Three Non-Collinear Points

Vector Form

$$[\vec{r}\vec{a}\vec{b}] + [\vec{r}\vec{a}\vec{c}] + [\vec{r}\vec{b}\vec{c}] = [\vec{a}\vec{b}\vec{c}] \text{ or } (\vec{r} - \vec{a}) \cdot (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) = 0$$

where, $\vec{a}, \vec{b}, \vec{c}$ are the position vector of three given noncollinear points through which the plane passes.

Cartesian Form

The equation of plane passing through three noncollinear points P with coordinates $(x_1, y_1, z_1), (x_2, y_2, z_2)$ & (x_3, y_3, z_3) is given as:

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$$

9. Intercept Form of The Equation of A Plane

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Where a, b, c are the intercepts made by the plane on x, y & z axes respectively.

10. Plane Passing Through The Intersection Of Two Given Planes

Vector Form

Equation of plane passing through the point of intersection of two planes

$\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is given as:

$$\vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = d_1 + \lambda d_2$$

Cartesian Form

$$\vec{n}_1 = A_1\hat{i} + B_1\hat{j} + C_1\hat{k}$$

$$\vec{n}_2 = A_2\hat{i} + B_2\hat{j} + C_2\hat{k} \text{ and } \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

therefore its cartesian equation is:

$$(A_1x + B_1y + C_1z - d_1) + \lambda(A_2x + B_2y + C_2z - d_2) = 0$$

12. Angle Between Two Planes

Vector Form: The angle between two planes $\vec{r} \cdot \vec{n}_1 = d_1$ & $\vec{r} \cdot \vec{n}_2 = d_2$ is given as:

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\| \|\vec{n}_2\|}$$

Cartesian Form: The angle between two planes

$a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is given as

$$\cos \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

- If two planes are perpendicular, then $\vec{n}_1 \cdot \vec{n}_2 = 0$ or $a_1a_2 + b_1b_2 + c_1c_2 = 0$

- If two planes are perpendicular, then $\vec{n}_1 = \lambda \vec{n}_2$ or $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

11. Coplanarity of Two Lines

Vector Form

Two lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ are coplanar, if $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$

Cartesian Form

Two lines $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ & $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$ are coplanar, if

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

13. Distance Of A Point From A Plane

Vector Form

Distance of a point with position vector $\vec{a}$ from a plane $\vec{r} \cdot \vec{n} = d$

is given as: $\frac{|\vec{a} \cdot \vec{n} - d|}{\|\vec{n}\|}$

Cartesian Form

Distance of a point (x_1, y_1, z_1) from a plane: $ax + by + cz = d$ is given as:

$$\frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

14. Angle Between A Line And A Plane

Vector Form

Angle between a line

$\vec{r} = \vec{a} + \lambda \vec{b}$ and a plane $\vec{r} \cdot \vec{n} = d$ is

$$\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{\|\vec{b}\| \|\vec{n}\|}$$

Cartesian Form

Angle between a line $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and a plane

$a_2x + b_2y + c_2z = d$ is given as:

$$\sin \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

• If line is perpendicular to the plane, then $\vec{n} = \lambda \vec{b}$ or $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

• If line is parallel to the plane, then $\vec{n} \cdot \vec{b} = 0$ or $a_1a_2 + b_1b_2 + c_1c_2 = 0$



A REASONABLE PROBABILITY IS THE ONLY CERTAINTY

-E.W.HOWE

PROBABILITY

1

Probability

If an experiment results in a total of $(m + n)$ outcomes which are equally likely and if 'm' outcomes are favourable to an event 'A' while 'n' are unfavorable, then the probability of occurrence of the event 'A' denoted by $P(A)$, is defined by

$$P(A) = \frac{m}{m+n} = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

3

Algebra of Events

- Event A or B or $A \cup B = \{w: w \in A \text{ or } w \in B\}$
- Event A and B or $A \cap B = \{w: w \in A \text{ and } w \in B\}$
- Event A but not B or $A - B = A \cap B'$

5

Probability of $A \cup B$, $A \cap B$ and $P(\text{not } A)$

If A and B are any two events, then

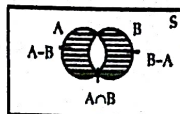
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

If A and B are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$

Probability of the event 'not A'

$$P(A') = P(\text{not } A) = 1 - P(A)$$



6

Conditional Probability

If E and F are two events associated with the sample space of a random experiment, the conditional probability of the event given that it has occurred is given as:

$$P(E/F) = \frac{P(E \cap F)}{P(F)} = \frac{n(E \cap F)}{n(F)}, P(F) \neq 0$$

Properties of Conditional Probability

1. Let E & F be events of sample space of an experiment, then we have

$$P(S/F) = P(F/F) = 1$$

2. If A and B are any two events of a sample space S & F is an event of such that

$$P(F) \neq 0, \text{ then } P((A \cup B)/F) = P(A/F) + P(B/F) - P((A \cap B)/F)$$

In particular if A and B are disjoint events, then

$$P((A \cup B)/F) = P(A/F) + P(B/F)$$

$$3. P(E/F) = 1 - P(E'/F)$$

7

Multiplication Theorem on Probability

For two events E & F associated with a sample space S, we have

$$P(E \cap F) = P(E)P(F/E) = P(F)P(E/F)$$

provided $P(E) \neq 0$ & $P(F) \neq 0$

The above result is known as Multiplication Rule of Probability.

8

Total Probability Theorem

If an event A can occur with one of the n mutually exclusive and exhaustive events $B_1, B_2, \dots, B_n$ and the probabilities $P(A/B_1), P(A/B_2), \dots, P(A/B_n)$ are known, then

$$P(A) = \sum_{i=1}^n P(B_i)P(A/B_i)$$

2

Random Experiment

An Experiment is called random experiment if it satisfies the following two conditions:

1. It has more than one possible outcome.
2. It is not possible to predict the outcome in advance.

Outcome: A possible result of a random experiment is called its outcome.

Sample Space: Set of all possible outcomes of a random experiment is called sample space. It is denoted by symbol 'S'.

4

Types of Events

1. **Impossible and Sure Event:** The empty set ϕ is called an Impossible event, where as the whole sample space 'S' is called 'Sure event'.

2. **Simple Event:** If an event has only one sample point of a sample space, it is called a 'simple event'.

3. **Compound Event:** If an event has more than one sample point, it is called a 'compound event'.

4. **Complementary Event:** Complement event to A = 'not A'

5. **Exhaustive Events:** Many events that together form sample space are called exhaustive events.

6. **Mutually Exclusive:** Events A & B are called mutually exclusive events if occurrence of any one of them excludes occurrence of other event, i.e. they cannot occur simultaneously.

7. **Mutually exclusive and exhaustive:** The events which are not mutually exclusive are known as compatible events or mutually non exclusive events. Mutually exclusive and exhaustive system of events: Let S be the sample space associated with a random experiment. Let $A_1, A_2, \dots, A_n$ be subsets of S such that

(i) $A_i \cap A_j = \phi$ for $i \neq j$ and

(ii) $A_1 \cup A_2 \cup \dots \cup A_n = S$

Then, the collection of events $A_1, A_2, \dots, A_n$ is said to form a mutually exclusive and exhaustive system of events.

8. **Independent Events**

(i) If E & F are independent, then

$$P(E \cap F) = P(F)P(E/F) = P(E), P(F) \neq 0, P(F/E) = P(F), P(E) \neq 0$$

(ii) Three events A, B & C are said to be mutually independent, if

$$P(A \cap B) = P(A)P(B), P(A \cap C) = P(A)P(C), P(B \cap C) = P(B)P(C) \text{ \& } P(A \cap B \cap C) = P(A)P(B)P(C)$$

If at least one of the above is not true for three given events, we say that the events are not independent.

9

Baye's Theorem

Partition of a Sample Space

A set of events $E_1, E_2, \dots, E_n$ is said to represent a partition of the sample space S if

$$(a) E_i \cap E_j = \phi, i \neq j, i, j = 1, 2, 3, \dots, n$$

$$(b) E_1 \cup E_2 \cup \dots \cup E_n = S$$

$$(c) P(E_i) > 0 \text{ for all } i = 1, 2, \dots, n$$

Theorem of Total Probability. Let $\{E_1, E_2, \dots, E_n\}$ be a partition of the sample space S

and suppose that each of the events $E_1, E_2, \dots, E_n$ has nonzero probability

of occurrence. Let A be any event associated with S then

$$P(A) = \sum_{i=1}^n P(E_i)P(A/E_i)$$

Baye's Theorem: If $E_1, E_2, \dots, E_n$ are non-empty events which constitute a partition of sample

space S & A is any event of non-zero probability.

$$P(E_i/A) = \frac{P(E_i)P(A/E_i)}{\sum_{j=1}^n P(E_j)P(A/E_j)} \text{ for any } i = 1, 2, 3, \dots, n$$

10 Random Variable & Its Probability Distributions

A random variable is a real valued function whose domain is the sample space of a random experiment. The probability distribution of a random variable

$$X: x_1 \quad x_2 \quad \dots \quad x_n$$

$$P(X): p_1 \quad p_2 \quad \dots \quad p_n$$

$$\text{where, } p_i > 0, \sum_{i=1}^n p_i = 1, i = 1, 2, \dots, n$$

The real numbers $x_1, x_2, \dots, x_n$ are the possible values

of the random variable X and $p_i (i = 1, 2, \dots, n)$

is the probability of the random variable i.e.,

$$P(X = x_i) = p_i$$

11 Mean Of A Random Variable

The mean (μ) of a random variable X is also called the expectation of X denoted by $E(X)$

$$E(X) = \mu = \sum_{i=1}^n x_i p_i$$

Here $x_1, x_2, \dots, x_n$ are possible values of random variable

X , occurring with probabilities $p_1, p_2, \dots, p_n$ respectively.

12 Variance Of Random Variable

Let X be a random variable whose possible values $x_1, x_2, \dots, x_n$ occur with probabilities $p(x_1), p(x_2), \dots, p(x_n)$ respectively. Also let

$\mu = E(X)$ be the mean of X then the variance of X is given as:

$$\text{Var}(X) \text{ or } \sigma_x^2 = \sum_{i=1}^n (x_i - \mu)^2 p(x_i) = E(X - \mu)^2 = E(X^2) - [E(X)]^2$$

The non-negative number $\sigma_x = \sqrt{\text{Var}(X)}$ is called the Standard

Deviation of random variable X

13 Bernoulli Trials & Binomial Distribution

Bernoulli Trials :

Trials of a random experiment are called Bernoulli trials, if they satisfy the following conditions:

- There should be a finite number of trials.
- The trials should be independent.
- Each trial has exactly two outcomes: success or failure.
- The probability of success remains same in each trial.

Binomial Distribution :

The probability distribution of number of successes in an experiment consisting of n Bernoulli trials may be obtained by the binomial expansion $(q + p)^n$ where p is probability of success in each trial and $p + q = 1$. Hence, this distribution (also called Binomial distribution $B(n, p)$) of number of successes X can be written as:

X	0	1	2	---	x	n
$P(x)$	${}^nC_0 q^n$	${}^nC_1 q^{n-1} p$	${}^nC_2 q^{n-2} p^2$		${}^nC_x q^{n-x} p^x$	${}^nC_n p^n$

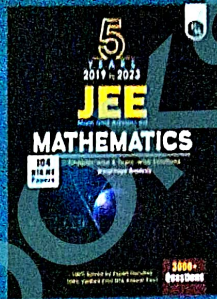
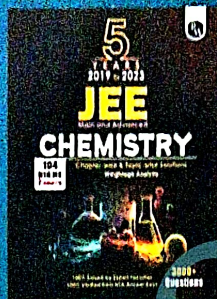
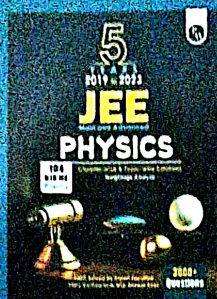
The probability of X successes $P(X = x)$ is also denoted by $P(x)$ is given as:

$$P(x) = {}^nC_x q^{n-x} p^x, x = 0, 1, \dots, n \quad (q = 1 - p)$$

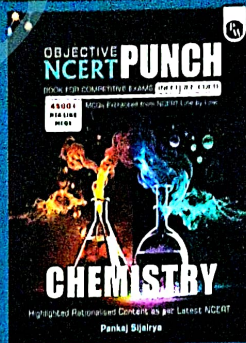
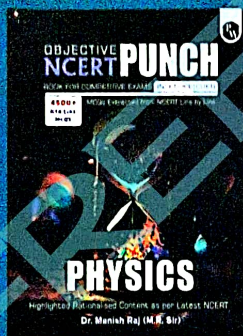
This $P(x)$ is called the probability function of the binomial distribution.

COMPLETE YOUR PREPARATION

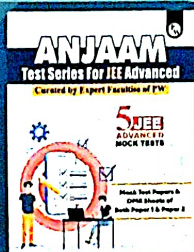
5 YEARS
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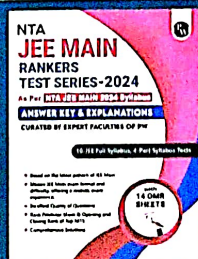
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